SMART LIGHTING & ZIGBEE MESH

Build plan · wiring · device map — Home Assistant "Beehive"

REV. AUG 13 2026
LOEWEN · WHITE HOUSE TN

THE STRATEGY — TWO JOBS, TWO BUDGETS

JOB 1 · LIGHT SWITCHES → WI-FI (KASA)

Kasa HS220/HS200 run **fully local** in Home Assistant through the `python-kasa` integration — no cloud, no hub, no account needed for on/off/dim. At ~$15 they do the lighting job for a fifth the price of a mesh-grade switch.

JOB 2 · MESH RANGE → ZIGBEE PLUGS

Mesh repeaters do **not** have to be switches. Every mains-powered Zigbee plug is a router. At $8–12 each they extend the mesh *and* replace the four vendor-locked Sylvania plugs that can never enter HA.

> **Why not a $46 mesh dimmer:** the switch was only being asked to repeat the mesh — a job a $10 plug does better. Buying them separately costs half as much and solves the Sylvania problem at the same time.

SHOPPING LIST

ITEM	FOR	QTY	EACH	TOTAL	STATUS
Kasa HS220 dimmer	Bedroom · Kitchen/Dining · Living room	3	$15	$0	2 ON HAND
Kasa HS220 dimmer	3rd room (only if a 3rd is wanted)	1	$15	$15	BUY
Kasa HS200 switch	Garage lights (non-dim)	1	$15	$15	BUY
Zigbee plug (4-pack)	Replace 4 Sylvania · mesh routers	4	~$10	$40	BUY
Zigbee plug — garage	Mesh relay through the garage wall	1	~$10	$10	BUY
Zigbee contact sensors	Garage door: CLOSED + FULLY-OPEN	2	~$12	$24	BUY
Zigbee coordinator dongle	Haozee CC2652P1 + USB extension	1	–	$0	ORDERED
Estimated total				**~$104**	

Prices are estimates — verify each Zigbee model against its Zigbee2MQTT device page before ordering (see Rules, page 3).

ROOM SCHEDULE

ROOM	FIXTURES	LOAD	SWITCH POSITIONS	DEVICE
Bedroom	9 × 12 W ProGreen LED	108 W	Door box only (2 toggles repurposed)	Kasa HS220
Kitchen + Dining	9 × 12 W (3-group + 6-group combined)	108 W	1 of 3 existing; 2 come out	Kasa HS220
Living Room	8 × 12 W LED	96 W	1 existing	Kasa HS220
Garage	8 × LED, not dimmed	96 W	**2 locations** — kitchen + garage	Kasa HS200 SEE NOTE

> **Open decision — garage two-location switching.** A single HS200 cannot serve two switch positions; the second position goes dead. Choose one: **(a)** HS210 matched kit so both positions stay live, or **(b)** single HS200 at the garage door and

> repurpose the kitchen position for something else — the same trick already used on the bedroom toggles. *Decide before ordering.*

> **Before first power-up:** install the Kasa app and **turn OFF automatic firmware updates before adding any switch.** TP-Link firmware has previously broken local control — and local control is the entire reason these are being used.

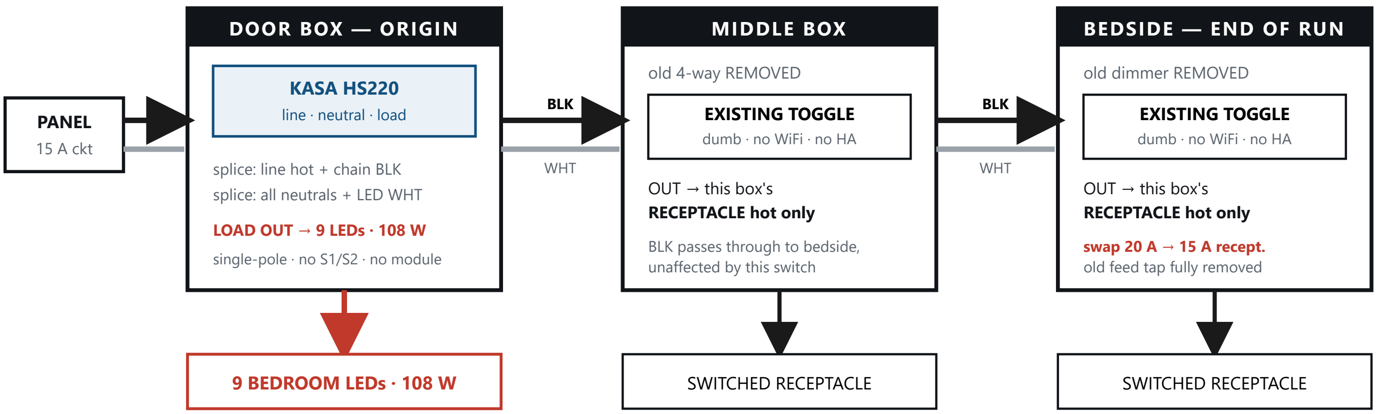
WIRING DIAGRAMS

Bedroom reversed-feed chain · kitchen consolidation · garage two-location

BLACK — constant hot · WHITE — neutral (unswitched, full length) · SWITCHED LEG — load out

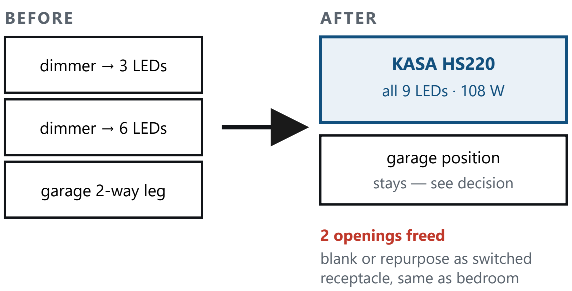
GROUND — bonded, never a circuit conductor

BEDROOM — REVERSED FEED, DOOR BOX IS THE ORIGIN

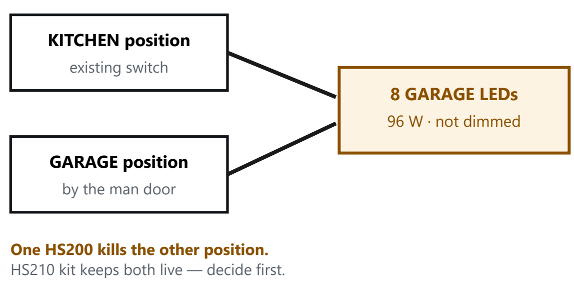


Bedside was originally the power origin — the feed is reversed so the door box originates. The two old toggles are not blanked off: each now controls only its own box's receptacle, entirely independent of the lights and of each other.

KITCHEN + DINING — 3 BOXES BECOME 1



GARAGE — THE TWO-LOCATION QUESTION

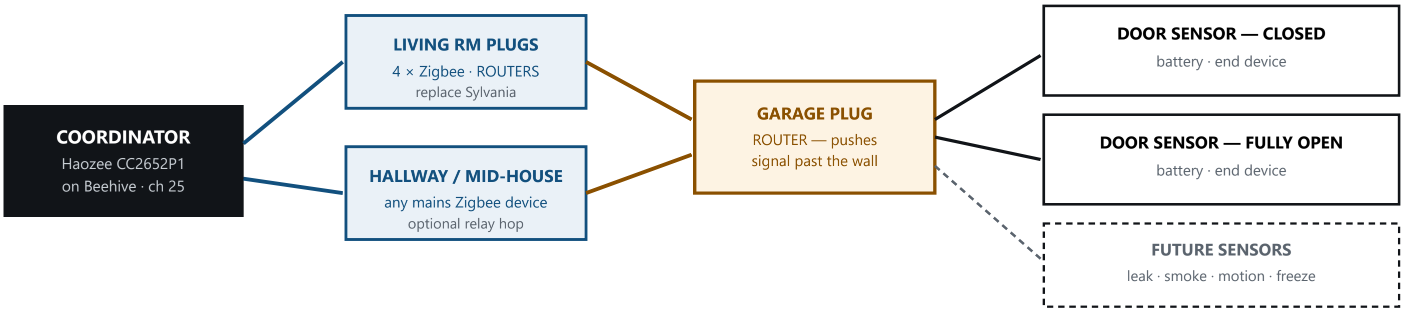


Before any box is opened: breaker OFF · verify dead at *all three* bedroom boxes · confirm a neutral is present at the kitchen and living-room boxes (assumed, not yet photo-verified) · confirm the switched conductor reaches the fixture junction before final connection. Grounds bond throughout and are never used as circuit conductors.

ZIGBEE MESH & DEVICE MAP

How range reaches the garage — and what every device on the network is

MESH BUILD — ROUTERS CARRY THE SIGNAL



Mains-powered = router — relays messages for everything else. **Battery-powered = end device** — talks only to its nearest router, never relays. This is why adding plugs strengthens the network and adding sensors does not.

> **The garage door gets two sensors, not one.** One reads CLOSED, the other reads FULLY OPEN. With both, Home Assistant can tell the three real states apart: **closed**, **fully open**, and **partially open** — the cracked-for-ventilation position, which a single sensor can never distinguish from wide open.

DEVICE MAP — WHAT EVERYTHING ACTUALLY IS

INFRASTRUCTURE

IP	WHAT IT IS
.254	AT&T BGW320-500 gateway
.66	**Beehive** — Home Assistant (fixed)
.194	**The beast** — main PC / CodeProject.AI
.196	RE200 — wired access point
.121	GaragePC — HP TouchSmart
.215	Fire TV (fixed)
.198	B-Hyve irrigation controller
.232	Mower sensor box (ESP32)

WI-FI NETWORKS

SSID	BAND	USE
Loewen301	2.4 · ch 1	all IoT + Kasa switches
Loewen301-5G	5 GHz	phones · laptops · TVs
LoewenGuest	2.4	guests only

Zigbee runs channel 25 — clear of Wi-Fi ch 1.

TUYA / SMART PLUGS (ALL IDENTIFIED AUG 13)

IP	DEVICE
.170	Garage fan socket
.224	Jeff's bed lamp socket
.171	Angela's bed lamp socket
.209	Hot-water pump socket
.195	Smart socket (spare)
.231	Sharky — robot vacuum
.199 .200 .202 .205	**4 × Sylvania plugs** — vendor-locked, cannot enter HA · **replace with Zigbee**

> **Settled — do not retry.** The Sylvania plugs are Tuya hardware locked to Sylvania's own app. Smart Life detects them and refuses; HA cannot reach them. Replacement is the fix.

BUYING RULES — LEARNED THE HARD WAY

- **"Zigbee compatible" is not enough.** Z-Wave and Zigbee are different radios — a Z-Wave switch will never talk to the Zigbee dongle, no matter what the box says.
- **Check the Zigbee2MQTT device page before buying.** One rejected dimmer was fully "supported" yet carried a warning that it *stops relaying messages for other devices* — useless for building a mesh.
- **Let cheap plugs carry the mesh**, not expensive switches. Same result, half the money.
- **Disable vendor auto-firmware-update** before first pairing on anything whose local control matters.

INSTALL CHECKLIST

PREP

- [ ] Kasa app installed
- [ ] Auto-update disabled
- [ ] Breaker OFF
- [ ] Verified dead at box
- [ ] Neutral confirmed

PER SWITCH

- [ ] Photo before removing
- [ ] Line vs load identified
- [ ] Kasa wired + mounted
- [ ] Power on · test manual
- [ ] Join Loewen301 (2.4)

HOME ASSISTANT

- [ ] Appears in HA
- [ ] Named by room
- [ ] Dim range checked
- [ ] Added to HCC app
- [ ] Logged in inventory